Null Graph

Graph Theory Explorer — Study Notes

Definition

A null graph (also called an empty graph) is a graph that contains a set of vertices but no edges connecting them. It is denoted as $N(n)$, where n is the number of vertices. Formally, $G = (V, E)$ where V is a non-empty set of vertices and $E = \varnothing$ (the empty set).

Key Properties

- The edge set E is always empty: $|E| = 0$.
- Every vertex has degree 0, since no vertex touches an edge.
- A null graph with a single vertex is also called a trivial graph.
- It is trivially disconnected whenever it has more than one vertex, since there is no path between any two distinct vertices.
- The sum of degrees of all vertices is 0, consistent with the Handshaking Lemma (sum of degrees = $2|E|$).

Example

Consider a graph with vertex set $V = \{A, B, C, D\}$ and edge set $E = \varnothing$. Here we have 4 isolated vertices A, B, C, and D with no edges drawn between any pair. Each vertex has degree 0.

Why It Matters

The null graph serves as the base case in many graph theory proofs and algorithms. It represents the 'empty' starting point before any relationships (edges) are added, and is often used to illustrate degree-sum arguments and to test edge-cases in graph algorithms (e.g., checking for connectivity or counting components).